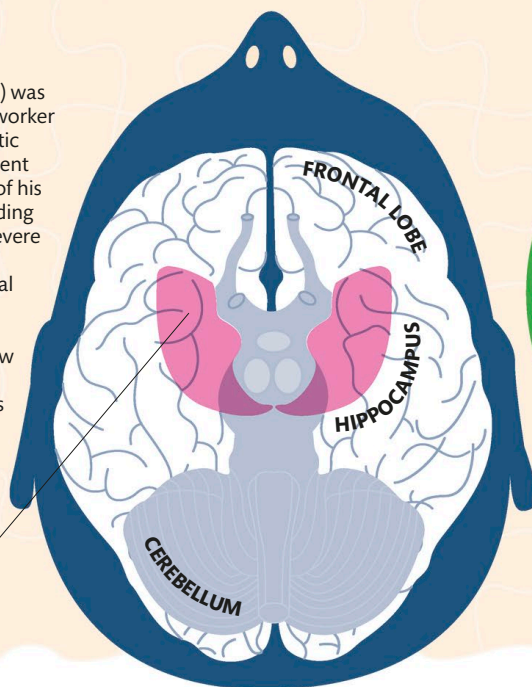


**A curious case**

Henry Molaison (1926–2008) was an American assembly line worker suffering from severe epileptic seizures. In 1953, he underwent surgery to remove sections of his medial temporal lobe, including both hippocampi, to treat severe epilepsy. This controlled his seizures, but he forgot several years before the surgery and developed anterograde amnesia. He could retain new declarative memories (see p.135) only for a few seconds but could learn new skills.



Large areas
of medial temporal lobe
removed from brain
in each hemisphere

VIEW FROM BELOW**WHAT IS
“SHELL SHOCK”?**

The expression was coined during World War I to describe an effect thought to be caused by the sound of exploding shells. Soldiers were, in fact, suffering from PTSD, brought on by the trauma of war.

Other memory problems

Many things affect memory, from short-term stress to life events, such as having children. Memory changes can be linked to changes in our neurochemistry. For example, cortisol is released when we worry and hormones surge in a pregnant woman around the time of birth. Lifestyle changes such as sleep deprivation also play a role.

CAUSE	EXPLANATION
Stress	Moderate, short-term stress can make it easier to form memories, but it becomes harder to recall facts you have already learned. This may explain why the feeling of “going blank” during an examination is so common.
Anxiety	Long-term or chronic stress, such as is experienced by people with anxiety disorders, can damage the hippocampus and other memory structures of the brain, causing memory problems.
Depression	Depression can impact the short-term memory and cause people to have difficulty recalling details of events they have experienced. Healthy people tend to remember positives better than negatives. In depression, this is reversed.
“Baby brain”	Pregnant women may experience mild decline in a range of cognitive abilities, although these are likely to be noticeable only to the women themselves. After the baby is born, sleep deprivation can worsen memory problems.

**POST-TRAUMATIC STRESS
DISORDER**

Normally when we store memories, the emotion fades over time, so we recall past events without reliving them. In post-traumatic stress disorder (PTSD), sufferers fail to dissociate memory from emotion, and intrusive memories bring the fear flooding back. These memories can be activated by sights or sounds, and often the patient is unaware of their triggers.

